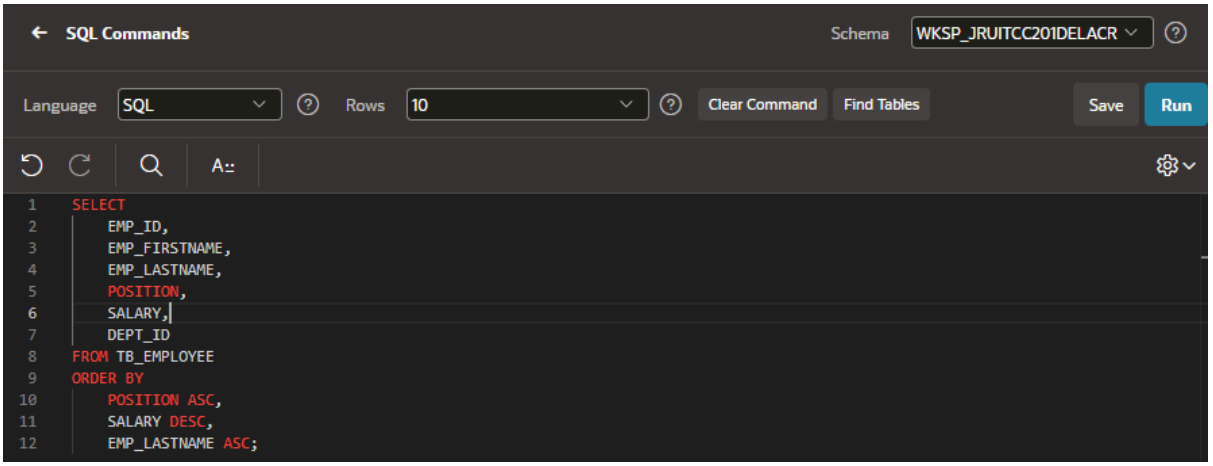
Dela Cruz, Eunice Rya Leigh
BSIT 201I

FINAL TERM ACTIVITY NO. 3 - SQL SORTING & JOINING TABLES PART 2

Multi-level Sorting



Results | Explain | Describe | Saved SQL | History

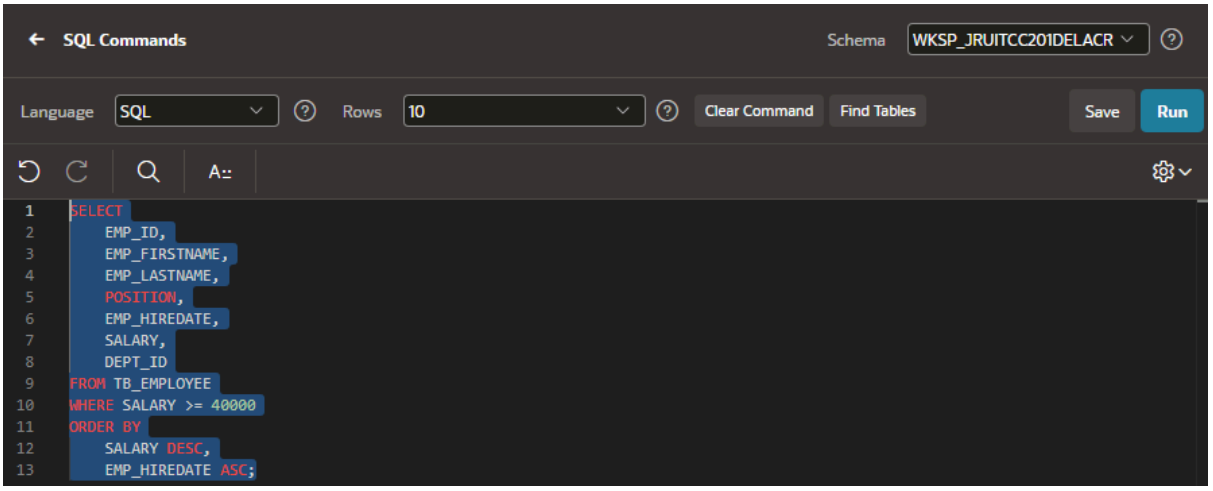
EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	SALARY	DEPT_ID
222224	Marris	Del Rosario	Analyst	53000	999990
222233	Arnie	Agtonon	Analyst	50000	777770
222232	Martin	Mesias	Analyst	48000	888880
222225	Alex	Causaren	Clerk	54000	999990
222228	Joseph	Trinidad	Clerk	19000	777770
222230	Jeffrey	Romero	Clerk	18500	888880
222226	Alley	Reyes	Manager	55000	555550
222221	Joanna	Mayuga	Manager	54000	888880
222223	Marissa	Abad	Manager	53000	666660
222222	Martha	Rosas	Manager	53000	777770

More than 10 rows available. Increase rows selector to view more rows.

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Salary Ranking

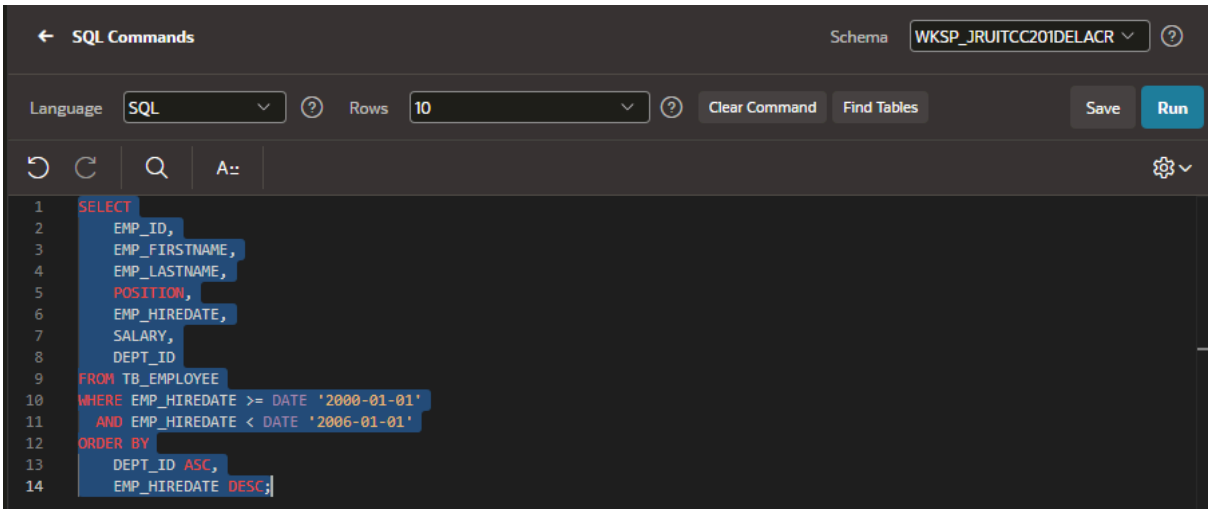


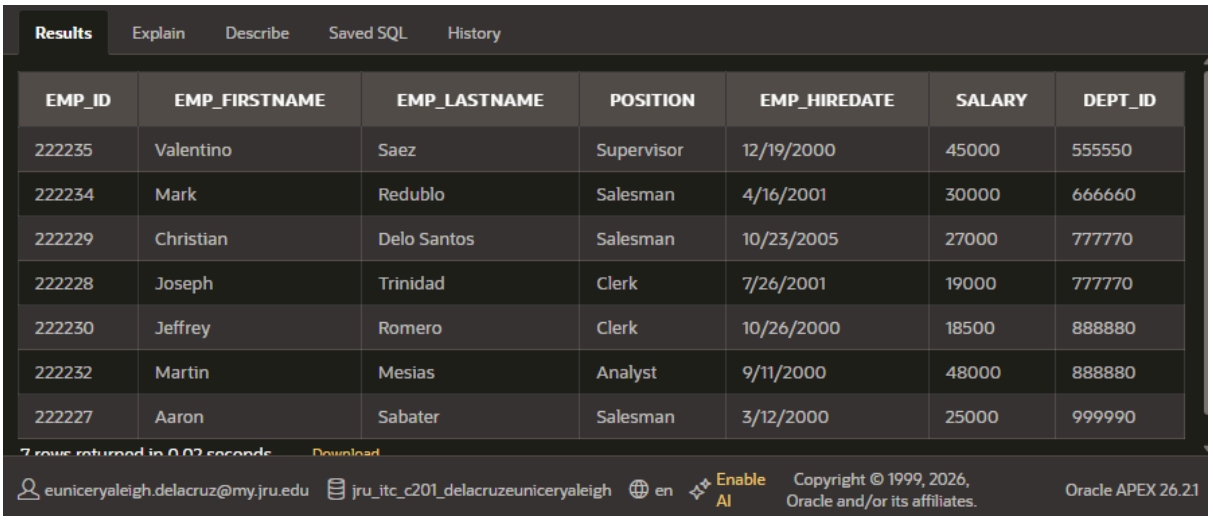
Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	EMP_HIREDATE	SALARY	DEPT_ID
222226	Alley	Reyes	Manager	4/3/1994	55000	555550
222225	Alex	Causaren	Clerk	8/2/1993	54000	999990
222221	Joanna	Mayuga	Manager	3/15/1996	54000	888880
222222	Martha	Rosas	Manager	5/30/1997	53000	777770
222223	Marissa	Abad	Manager	2/25/1999	53000	666660
222220	Rashley	Torres	Manager	2/25/1999	53000	999990
222224	Marris	Del Rosario	Analyst	9/11/1999	53000	999990
222233	Arnie	Agtonon	Analyst	5/21/1994	50000	777770
222232	Martin	Mesias	Analyst	9/11/2000	48000	888880
222235	Valentino	Saez	Supervisor	12/19/2000	45000	555550

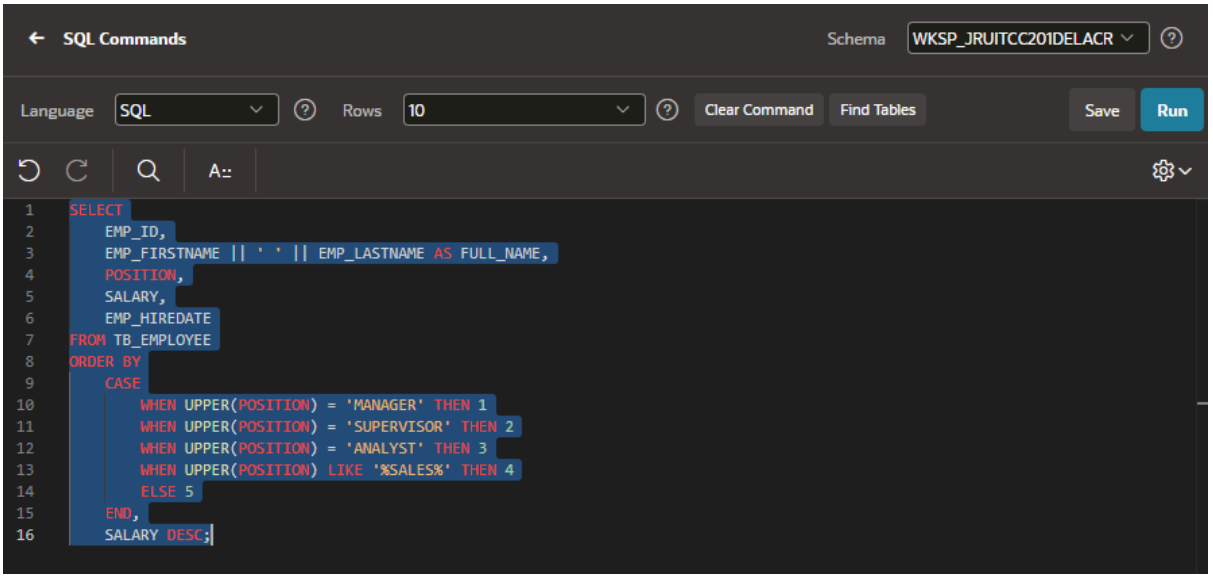
10 rows returned in 0.02 seconds Download

Department and Hire-Date Ordering





Conditional Sorting




```sql
SELECT
    EMP_ID,
    EMP_FIRSTNAME || ' ' || EMP_LASTNAME AS FULL_NAME,
    POSITION,
    SALARY,
    EMP_HIREDATE
FROM TB_EMPLOYEE
ORDER BY
    CASE
        WHEN UPPER(POSITION) = 'MANAGER' THEN 1
        WHEN UPPER(POSITION) = 'SUPERVISOR' THEN 2
        WHEN UPPER(POSITION) = 'ANALYST' THEN 3
        WHEN UPPER(POSITION) LIKE '%SALES%' THEN 4
        ELSE 5
    END,
    SALARY DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	FULL_NAME	POSITION	SALARY	EMP_HIREDATE
222226	Alley Reyes	Manager	55000	4/3/1994
222221	Joanna Mayuga	Manager	54000	3/15/1996
222222	Martha Rosas	Manager	53000	5/30/1997
222223	Marissa Abad	Manager	53000	2/25/1999
222220	Rashley Torres	Manager	53000	2/25/1999
222235	Valentino Saez	Supervisor	45000	12/19/2000
222224	Marris Del Rosario	Analyst	53000	9/11/1999
222233	Arnie Agtonon	Analyst	50000	5/21/1994
222232	Martin Mesias	Analyst	48000	9/11/2000
222231	Martheena Herrera	Salesman	30000	11/22/1999

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.02 seconds Download

Part B. Joining TB_EMPLOYEE and TB_DEPARTMENT
Basic INNER JOIN

SQL Commands — Schema: WKSP_JRUITCC201DELACR

Language: SQL — Rows: 10 — Clear Command — Find Tables — Save — Run

```sql
SELECT
    E.EMP_ID,
    E.EMP_FIRSTNAME || ' ' || E.EMP_LASTNAME AS FULL_NAME,
    E.POSITION,
    E.SALARY,
    D.DEPT_NAME,
    D.LOCATION
FROM TB_EMPLOYEE E
INNER JOIN TB_DEPARTMENT D
    ON E.DEPT_ID = D.DEPT_ID
ORDER BY
    D.DEPT_NAME ASC,
    E.SALARY DESC;
```

Results | Explain | Describe | Saved SQL | History

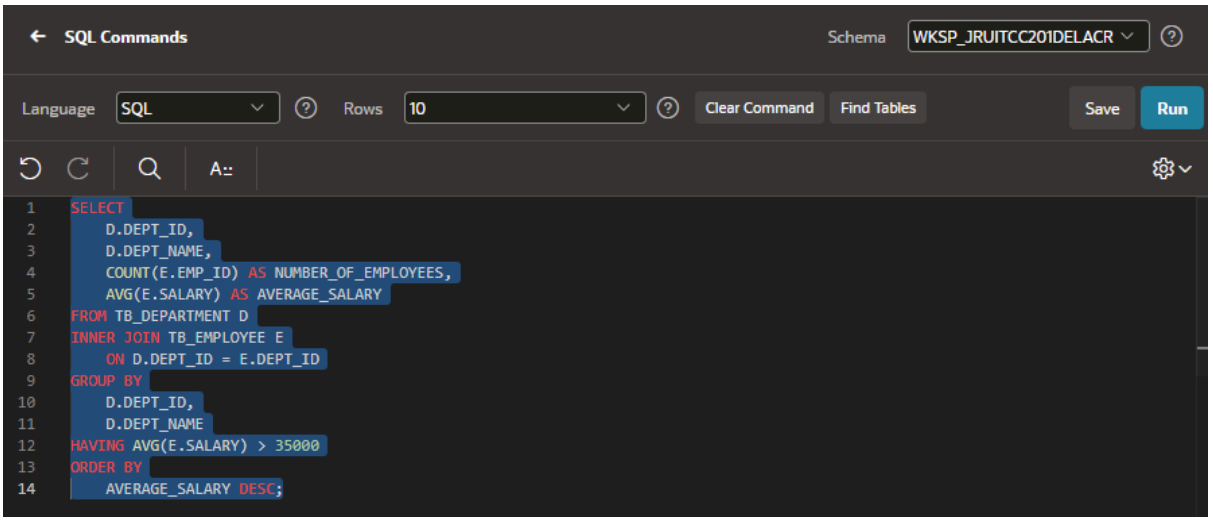
EMP_ID	FULL_NAME	POSITION	SALARY	DEPT_NAME	LOCATION
222225	Alex Causaren	Clerk	54000	Accounting	3F Rm 301
222220	Rashley Torres	Manager	53000	Accounting	3F Rm 301
222224	Marris Del Rosario	Analyst	53000	Accounting	3F Rm 301
222227	Aaron Sabater	Salesman	25000	Accounting	3F Rm 301
222221	Joanna Mayuga	Manager	54000	Marketing	4F Rm 411
222232	Martin Mesias	Analyst	48000	Marketing	4F Rm 411
222231	Martheena Herrera	Salesman	30000	Marketing	4F Rm 411
222230	Jeffrey Romero	Clerk	18500	Marketing	4F Rm 411
222226	Alley Reyes	Manager	55000	Payroll	2F Rm 201
222235	Valentino Saez	Supervisor	45000	Payroll	2F Rm 201

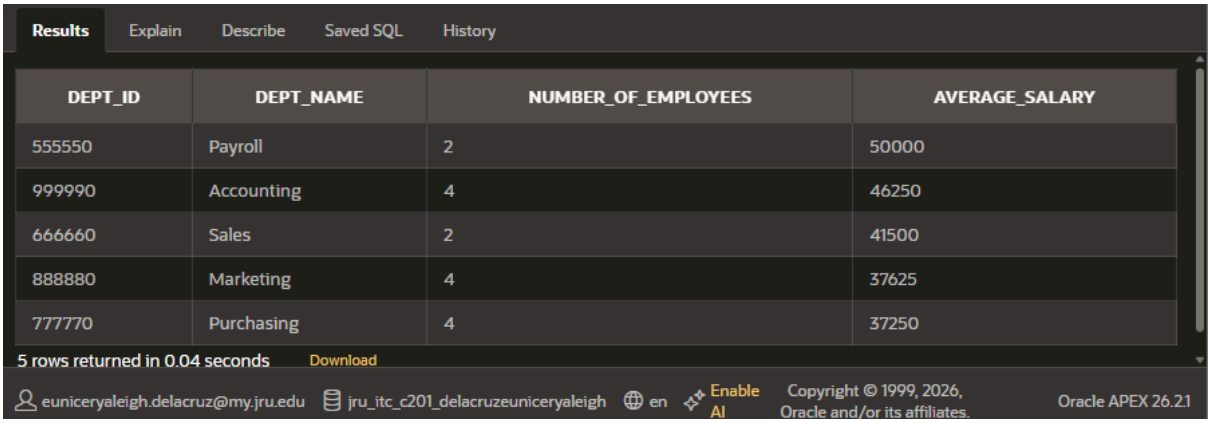
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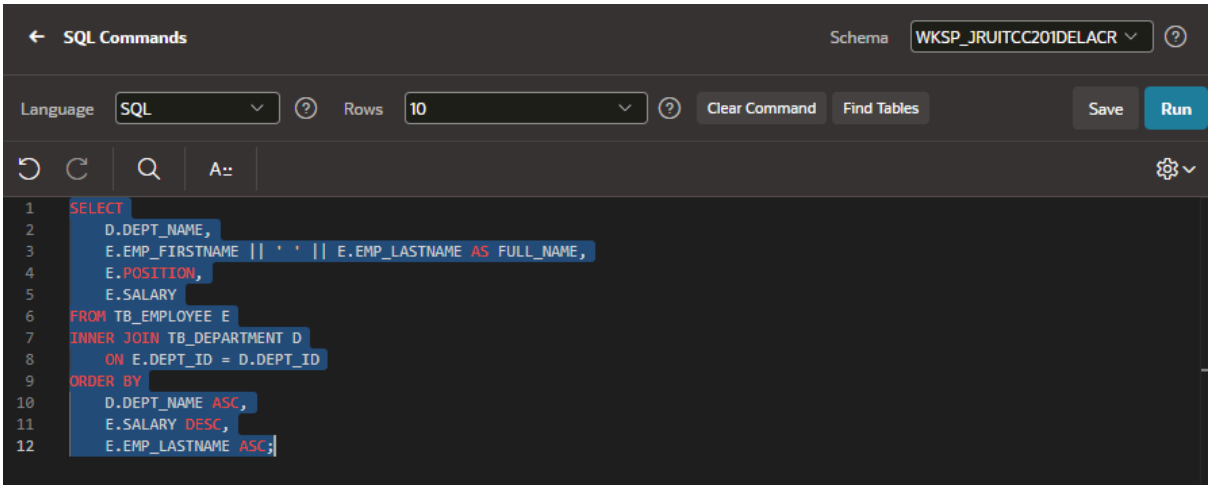
Department Salary Analysis





DEPT_ID	DEPT_NAME	NUMBER_OF_EMPLOYEES	AVERAGE_SALARY
555550	Payroll	2	50000
999990	Accounting	4	46250
666660	Sales	2	41500
888880	Marketing	4	37625
777770	Purchasing	4	37250

Department-Specific Employee Ranking




```sql
SELECT
    D.DEPT_NAME,
    E.EMP_FIRSTNAME || ' ' || E.EMP_LASTNAME AS FULL_NAME,
    E.POSITION,
    E.SALARY
FROM TB_EMPLOYEE E
INNER JOIN TB_DEPARTMENT D
    ON E.DEPT_ID = D.DEPT_ID
ORDER BY
    D.DEPT_NAME ASC,
    E.SALARY DESC,
    E.EMP_LASTNAME ASC;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	FULL_NAME	POSITION	SALARY
Accounting	Alex Causaren	Clerk	54000
Accounting	Marris Del Rosario	Analyst	53000
Accounting	Rashley Torres	Manager	53000
Accounting	Aaron Sabater	Salesman	25000
Marketing	Joanna Mayuga	Manager	54000
Marketing	Martin Mesias	Analyst	48000
Marketing	Martheena Herrera	Salesman	30000
Marketing	Jeffrey Romero	Clerk	18500
Payroll	Alley Reyes	Manager	55000
Payroll	Valentino Saez	Supervisor	45000

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds Download

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LEFT JOIN and Unmatched Departments

SQL Commands

Schema: WKSP_JRUITCC201DELACR

Language: SQL | Rows: 10 | Clear Command | Find Tables | Save | Run

```sql
SELECT
    D.DEPT_ID,
    D.DEPT_NAME,
    E.EMP_ID,
    E.EMP_FIRSTNAME || ' ' || E.EMP_LASTNAME AS FULL_NAME,
    E.SALARY
FROM TB_DEPARTMENT D
LEFT JOIN TB_EMPLOYEE E
    ON D.DEPT_ID = E.DEPT_ID
ORDER BY
    D.DEPT_ID ASC,
    E.EMP_LASTNAME ASC NULLS LAST;
```

Results | Explain | Describe | Saved SQL | History

DEPT_ID	DEPT_NAME	EMP_ID	FULL_NAME	SALARY
555550	Payroll	222226	Alley Reyes	55000
555550	Payroll	222235	Valentino Saez	45000
666660	Sales	222223	Marissa Abad	53000
666660	Sales	222234	Mark Redublo	30000
777770	Purchasing	222233	Arnie Agtonon	50000
777770	Purchasing	222229	Christian Delo Santos	27000
777770	Purchasing	222222	Martha Rosas	53000
777770	Purchasing	222228	Joseph Trinidad	19000
888880	Marketing	222231	Martheena Herrera	30000
888880	Marketing	222221	Joanna Mayuga	54000

More than 10 rows available. Increase rows selector to view more rows.

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Employees Earning Above Their Department Average

SQL Commands

Schema: WKSP_JRUITCC201DELACR

Language: SQL | Rows: 10 | Clear Command | Find Tables | Save | Run

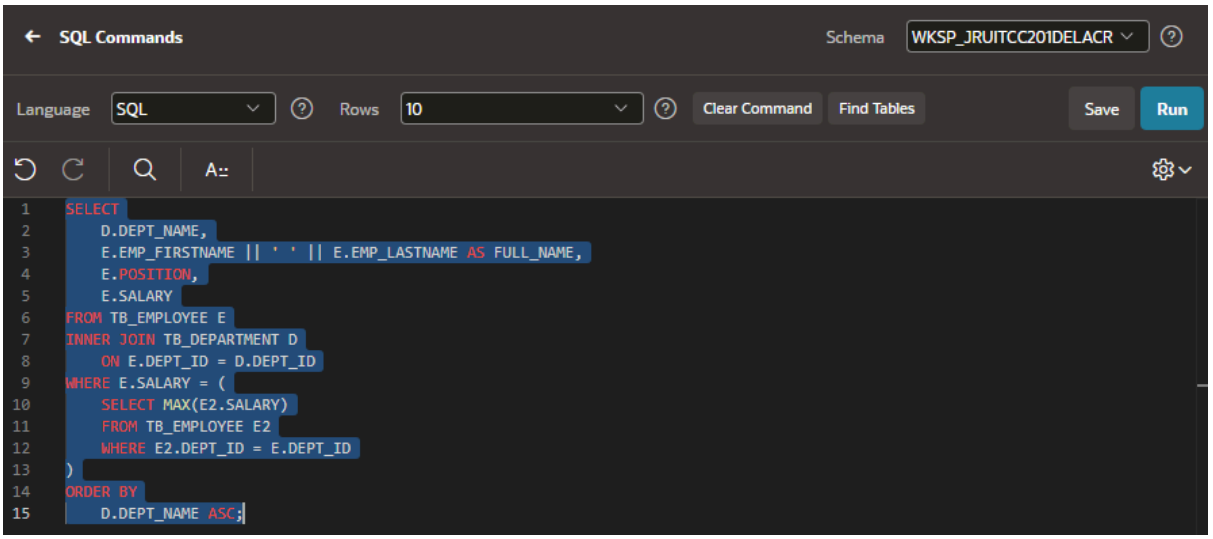
```sql
SELECT
    E.EMP_ID,
    E.EMP_FIRSTNAME || ' ' || E.EMP_LASTNAME AS FULL_NAME,
    E.SALARY,
    D.DEPT_NAME,
    (
        SELECT AVG(E2.SALARY)
        FROM TB_EMPLOYEE E2
        WHERE E2.DEPT_ID = E.DEPT_ID
    ) AS DEPARTMENT_AVERAGE
FROM TB_EMPLOYEE E
INNER JOIN TB_DEPARTMENT D
    ON E.DEPT_ID = D.DEPT_ID
WHERE E.SALARY > (
    SELECT AVG(E2.SALARY)
    FROM TB_EMPLOYEE E2
    WHERE E2.DEPT_ID = E.DEPT_ID
)
ORDER BY
    D.DEPT_NAME ASC,
    E.SALARY DESC;
```

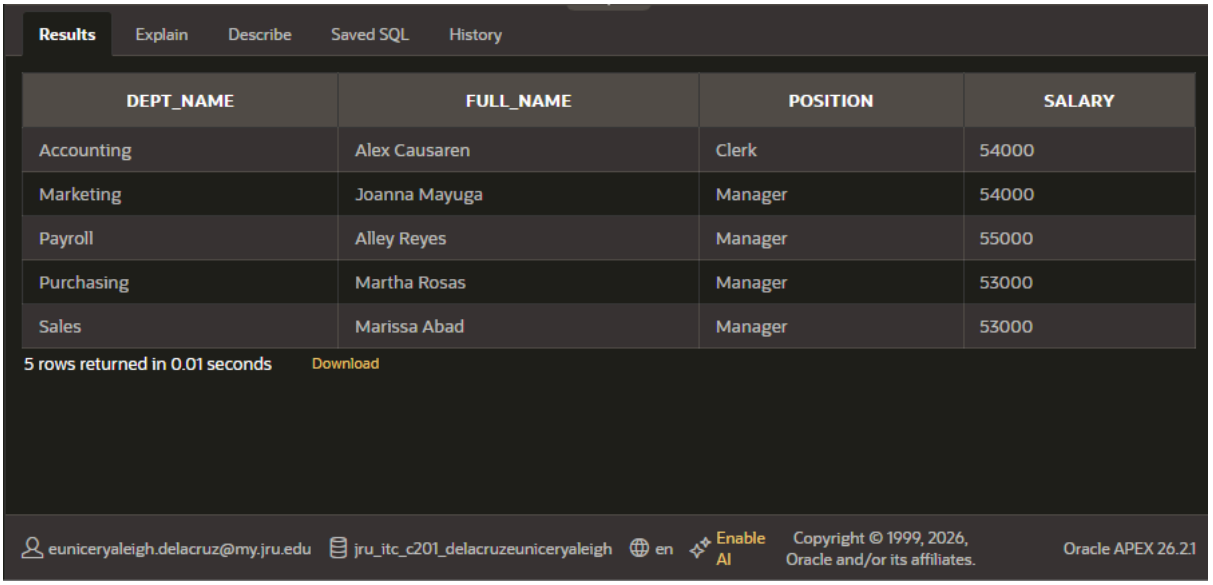
Results | Explain | Describe | Saved SQL | History

EMP_ID	FULL_NAME	SALARY	DEPT_NAME	DEPARTMENT_AVERAGE
222225	Alex Causaren	54000	Accounting	46250
222220	Rashley Torres	53000	Accounting	46250
222224	Marris Del Rosario	53000	Accounting	46250
222221	Joanna Mayuga	54000	Marketing	37625
222232	Martin Mesias	48000	Marketing	37625
222226	Alley Reyes	55000	Payroll	50000
222222	Martha Rosas	53000	Purchasing	37250
222233	Arnie Agtonon	50000	Purchasing	37250
222223	Marissa Abad	53000	Sales	41500

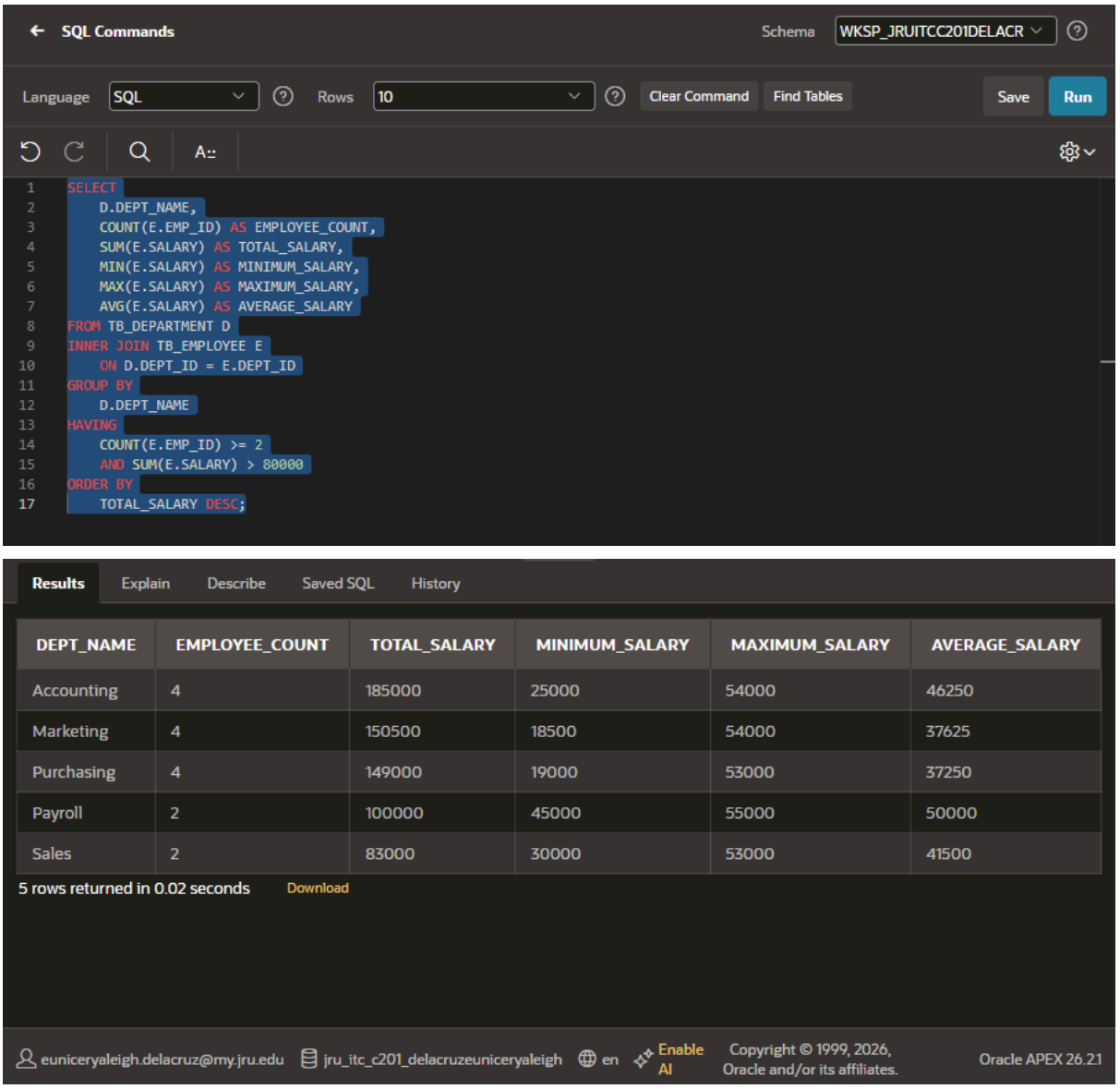
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Highest-Paid Employee per Department





Department Salary Distribution



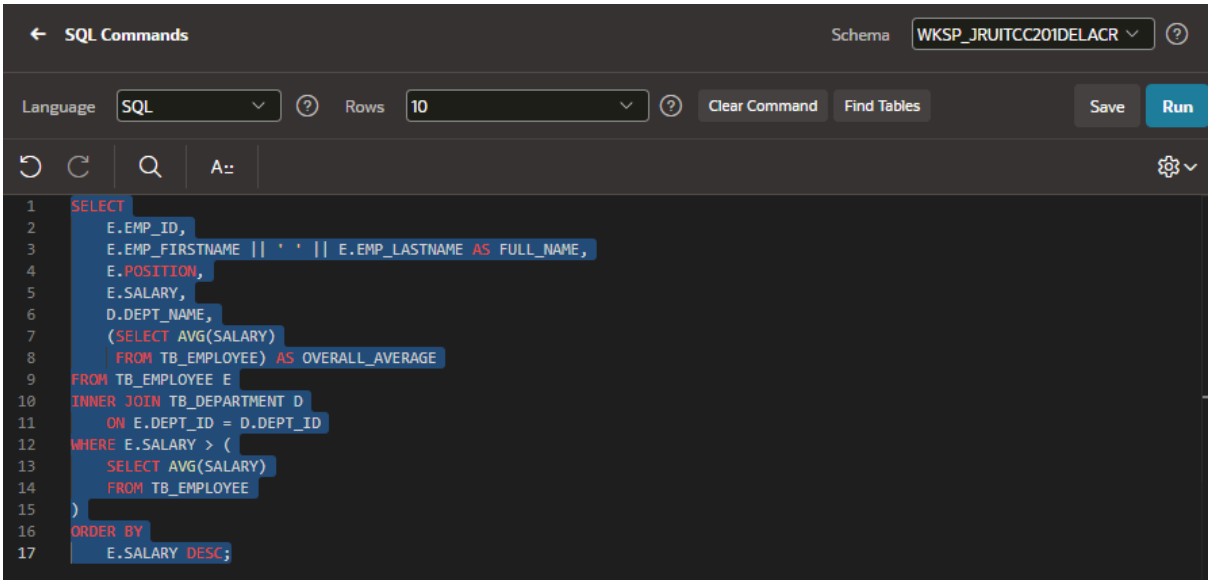
Results | Explain | Describe | Saved SQL | History

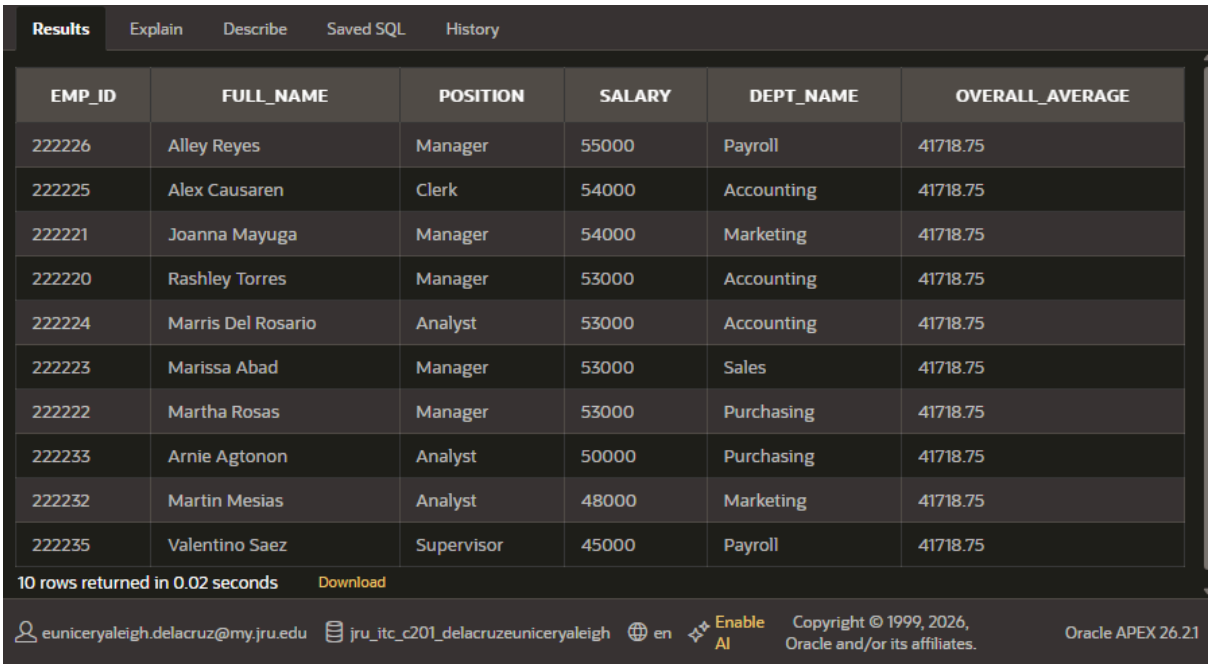
DEPT_NAME	EMPLOYEE_COUNT	TOTAL_SALARY	MINIMUM_SALARY	MAXIMUM_SALARY	AVERAGE_SALARY
Accounting	4	185000	25000	54000	46250
Marketing	4	150500	18500	54000	37625
Purchasing	4	149000	19000	53000	37250
Payroll	2	100000	45000	55000	50000
Sales	2	83000	30000	53000	41500

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Employees with Salaries Above the Company Average





Department Comparison

SQL Commands

Schema: WKSP_JRUITCC201DELACR

Language: SQL | Rows: 10 | Clear Command | Find Tables | Save | Run

```
SELECT
    D.DEPT_NAME,
    COUNT(E.EMP_ID) AS EMPLOYEE_COUNT,
    AVG(E.SALARY) AS DEPARTMENT_AVERAGE,
    (SELECT AVG(SALARY)
     FROM TB_EMPLOYEE) AS COMPANY_AVERAGE
FROM TB_DEPARTMENT D
INNER JOIN TB_EMPLOYEE E
    ON D.DEPT_ID = E.DEPT_ID
GROUP BY
    D.DEPT_ID,
    D.DEPT_NAME
HAVING AVG(E.SALARY) > (
    SELECT AVG(SALARY)
    FROM TB_EMPLOYEE
)
ORDER BY
    AVG(E.SALARY) - (
        SELECT AVG(SALARY)
        FROM TB_EMPLOYEE
    ) DESC;
```

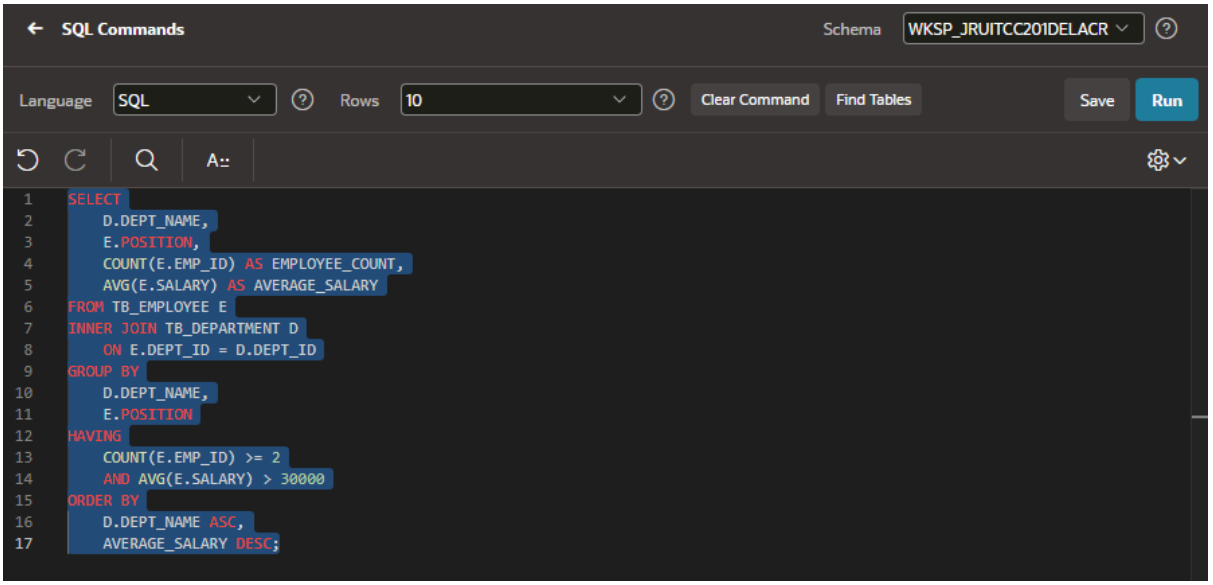
Results | Explain | Describe | Saved SQL | History

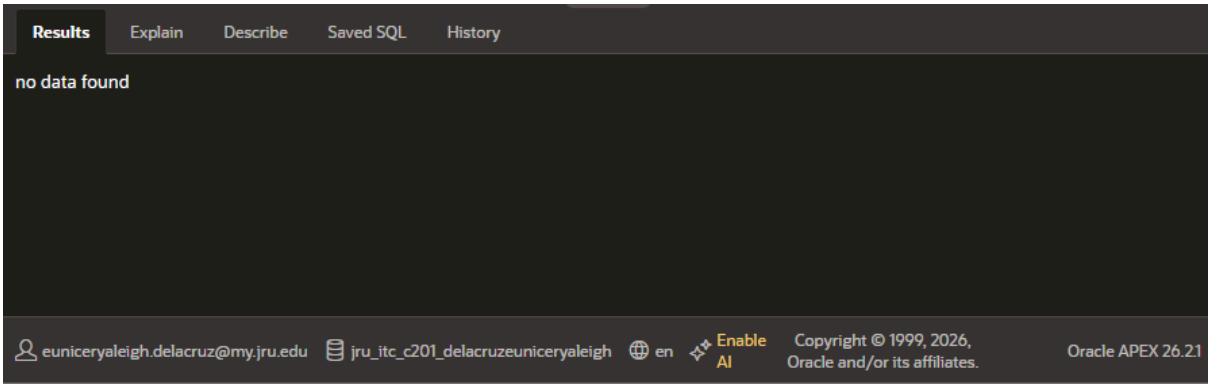
DEPT_NAME	EMPLOYEE_COUNT	DEPARTMENT_AVERAGE	COMPANY_AVERAGE
Payroll	2	50000	41718.75
Accounting	4	46250	41718.75

2 rows returned in 0.02 seconds Download

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Position-Based Salary Analysis





Advanced Conditional Join and Sorting

SQL Commands

Schema: WKSP_JRUITCC201DELACR

Language: SQL | Rows: 10 | Clear Command | Find Tables | Save | Run

```sql
SELECT
    E.EMP_ID,
    E.EMP_FIRSTNAME || ' ' || E.EMP_LASTNAME AS FULL_NAME,
    E.POSITION,
    E.EMP_HIREDATE,
    E.SALARY,
    D.DEPT_NAME,
    D.LOCATION
FROM TB_EMPLOYEE E
INNER JOIN TB_DEPARTMENT D
    ON E.DEPT_ID = D.DEPT_ID
WHERE
    E.SALARY > 45000
    OR (
        UPPER(E.POSITION) = 'MANAGER'
        AND E.SALARY >= 40000
    )
    OR (
        E.EMP_HIREDATE < DATE '2000-01-01'
        AND E.SALARY > 30000
    )
ORDER BY
    D.DEPT_NAME ASC,
    E.SALARY DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	FULL_NAME	POSITION	EMP_HIREDATE	SALARY	DEPT_NAME	LOCATION
222225	Alex Causaren	Clerk	8/2/1993	54000	Accounting	3F Rm 301
222220	Rashley Torres	Manager	2/25/1999	53000	Accounting	3F Rm 301
222224	Marris Del Rosario	Analyst	9/11/1999	53000	Accounting	3F Rm 301
222221	Joanna Mayuga	Manager	3/15/1996	54000	Marketing	4F Rm 411
222232	Martin Mesias	Analyst	9/11/2000	48000	Marketing	4F Rm 411
222226	Alley Reyes	Manager	4/3/1994	55000	Payroll	2F Rm 201
222222	Martha Rosas	Manager	5/30/1997	53000	Purchasing	4F Rm 413
222233	Arnie Agtonon	Analyst	5/21/1994	50000	Purchasing	4F Rm 413
222223	Marissa Abad	Manager	2/25/1999	53000	Sales	3F Rm 312

9 rows returned in 0.02 seconds Download

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